

Incidence and Prevalence of T1D in children and adolescents in Belgium pre- and post- COVID-19

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BACKGROUND

The global **incidence** and **prevalence** of type 1 diabetes (**T1D**) in children and adolescents is **rising worldwide**, and the **COVID-19 pandemic** has raised concerns about a potential influence on this trend

➡ This study quantified the incidence and prevalence of T1D in Belgian youth and assessed the impact of the COVID-19 pandemic on these metrics

METHODS

 Initiative for Quality Improvement and Epidemiology in Children and Adolescents with Diabetes (**IQECAD**) data (**2013-2023**)

 T1D **incidence** and **prevalence** assessed for **Belgian youth** (<19 years), stratified by **age** (0-5 years, 6-10 years, 11-18 years) and **sex**

 Population data obtained from the Belgian Statistical Office **STATBEL**

 Incidence rate ratios (**IRR**) were calculated using **Poisson regression**, comparing **pre- and post-COVID-19** (reference 2019)

CONCLUSION

- **Prevalence increased** from 2013 (0.11%) to 2023 (0.15%)
- **Incidence increased** from 2013 onwards - significant **rise** in **2021** (IRR=1.22, p<0.01), **decrease** in **2023** (IRR=0.85, p<0.05)
- Similar trend across **both** sexes and age groups
- **Children** diagnosed between **6 and 10 years** consistently showed the **highest incidence** of T1D

RESULTS

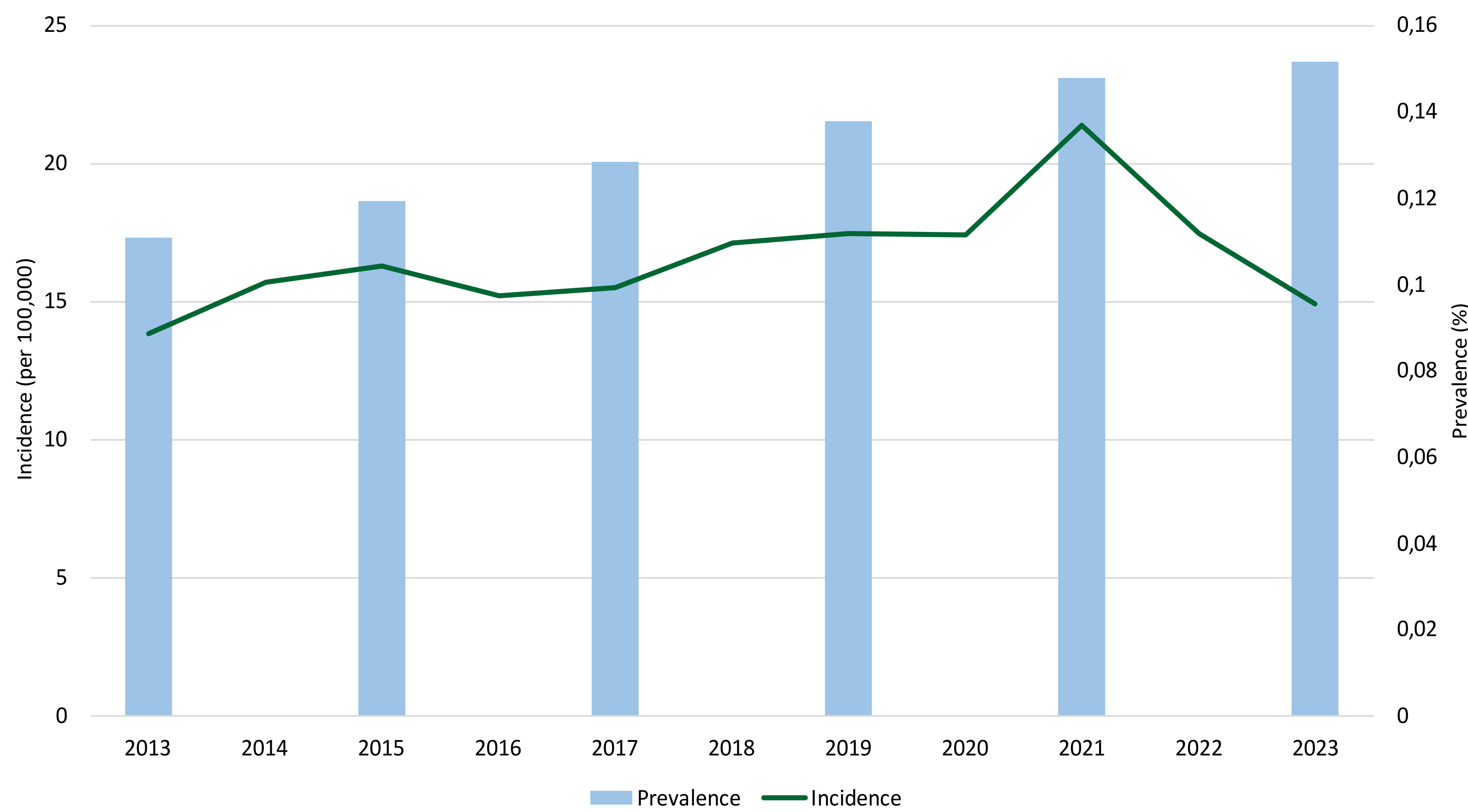


Figure 1: Prevalence and incidence of T1D in Belgian total youth population. Prevalence is calculated during each audit year (every two years – blue bars), and incidence is calculated for every year (green line).



Figure 2: Incidence of T1D (per 100,000) in Belgian youth by age group and sex. *p<0.05; **p<0.01; ***p<0.001, reference year = 2019

Table 1: Study population characteristics per audit.

Audit year	2013	2015	2017	2019	2021	2023	Trend since 2013, p value
Number of patients in study population	2653	2880	3123	3365	3618	3738	
Diabetes prevalence (%)	0.11%	0.12%	0.13%	0.14%	0.15%	0.15%	***
Boys, n (%)	1399 (52.7)	1511 (52.5)	1627 (52.1)	1750 (52.0)	1887 (52.2)	1935 (51.8)	
Age (years)	13.3 ± 4.0	13.4 ± 3.9	13.5 ± 3.9	13.5 ± 3.9	13.5 ± 4.0	13.6 ± 4.0	***
Diabetes duration (years)	5.4 ± 3.9	5.4 ± 3.9	5.5 ± 3.9	5.5 ± 3.9	5.3 ± 3.9	5.4 ± 3.9	***
HbA1c (%)	7.7 ± 1.2	7.7 ± 1.2	7.6 ± 1.1	7.7 ± 1.1	7.6 ± 1.1	7.6 ± 1.2	***
Celiac disease, n (%)	83 (3.4)	113 (4.4)	127 (4.2)	136 (4.2)	160 (4.7)	167 (4.7)	
Thyroid auto-immunity, n (%)	207 (8.6)	200 (7.8)	273 (9.1)	268 (8.2)	258 (7.6)	323 (9.0)	***

*p<0.05; **p<0.01; ***p<0.001

Table 2: Incidence of T1D in Belgian total youth population.

Year of diagnosis	New cases	Population	Incidence* 100.000	IRR	IRR
				[95% confidence interval]	[95% confidence interval]
				Reference year: 2013	Reference year: 2019
Total population					
2013	331	2,394,373	13.8	-	-
[...]					
2019	426	2,439,065	17.5	1.2634 [1.09 - 1.46] **	-
2020	425	2,437,567	17.4	1.2612 [1.09 - 1.46] **	0.9983 [0.87 - 1.14]
2021	523	2,446,111	21.4	1.5466 [1.35 - 1.77] ***	1.2242 [1.08 -1.39] **
2022	431	2,468,685	17.5	1.2629 [1.09 - 1.46] **	0.9996 [0.87 -1.14]
2023	368	2,465,820	14.9	1.0796 [0.93 - 1.25]	0.8545 [0.74 -0.98] *

*p<0.05; **p<0.01; ***p<0.001, reference year = 2019